

Stepped Vacuum Control Without PID

Scaling, percent-error bands, timed speed correction, and a complete RSLogix 500 implementation

Based on Project 8: 0–30 inHg sensor, 0–16383 raw scaling, 15 inHg setpoint, and one VFD correction every 10 seconds.

Core idea

Every 10 seconds, the PLC compares measured vacuum with the setpoint. The magnitude of the percentage error selects a correction of 0, 20, 60, or 100 raw VFD counts. The sign of the error selects whether that correction is added to or subtracted from the pump-speed command.

The counterintuitive direction rule

For this project, a **lower inHg value means more vacuum**. Therefore, if the measured value is above the setpoint, the receiver has insufficient vacuum and pump speed must increase. If the measured value is below the setpoint, pump speed must decrease.

1 Process Specification and Address Map

The PLC receives a scaled analog signal from the vacuum sensor and writes a raw speed reference to the pump's variable-frequency drive (VFD). The required controller is a discrete, stepped approximation to integral action rather than a PID loop.

Address	Purpose
N7:0	Raw vacuum-sensor input, nominally 0–16383 counts
N7:1	Raw pump/VFD speed command, limited to 0–16383 counts
N7:2	Vacuum setpoint in inHg; project test value is 15
N7:3	Scaled measured vacuum in inHg
N7:4	Signed control error, measured minus setpoint
N7:5	Absolute error magnitude
N7:6	Error magnitude as an integer percentage of setpoint
T4:0	Ten-second interval timer

1.1 Required correction schedule

Absolute percentage error	Correction	Action
$0\% \leq e_{\%} \leq 4\%$	0	Hold the existing VFD command
$4\% < e_{\%} \leq 15\%$	20	Add or subtract 20 counts
$15\% < e_{\%} \leq 30\%$	60	Add or subtract 60 counts
$e_{\%} > 30\%$	100	Add or subtract 100 counts

The use of strict lower bounds and inclusive upper bounds prevents two correction rungs from being true at the same boundary.

2 Controller Mathematics

2.1 Scale the sensor input

The input module uses PID-style raw scaling: 0 counts represents 0 inHg and 16383 counts represents 30 inHg. The linear conversion is

$$\text{Measured vacuum} = N7:3 = \frac{N7:0 \times 30}{16383}. \quad (1)$$

For the first emulator test,

$$N7:0 = 8192 \implies N7:3 = \frac{8192 \times 30}{16383} \approx 15.00 \text{ inHg}.$$

2.2 Calculate signed and absolute error

Define the signed error as

$$N7:4 = N7:3 - N7:2. \quad (2)$$

Its sign supplies the direction:

- $N7:4 > 0$: measured inHg is too high, vacuum is insufficient, so **increase** speed.
- $N7:4 < 0$: measured inHg is too low, vacuum is excessive, so **decrease** speed.

The magnitude is

$$N7:5 = |N7:4|. \quad (3)$$

2.3 Convert error to percent of setpoint

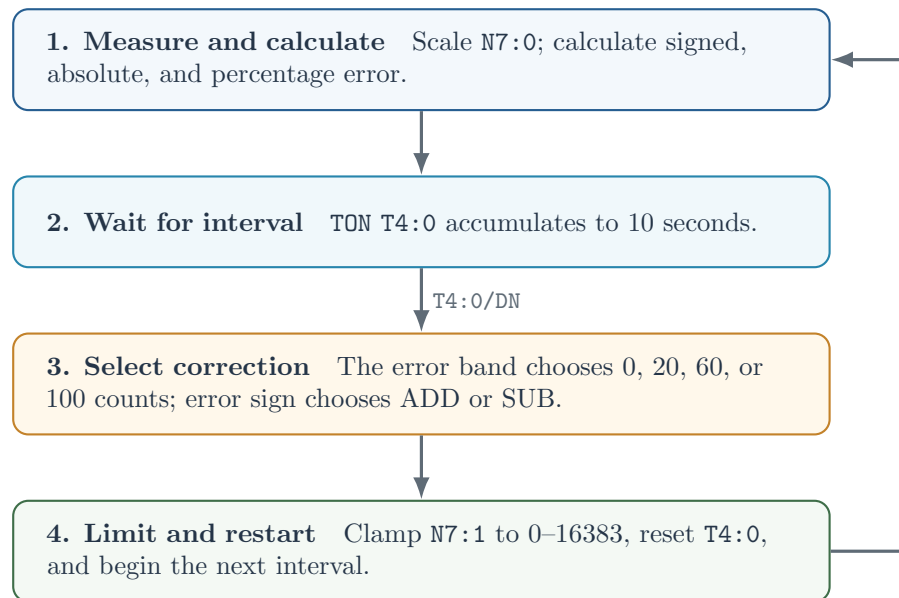
$$N7:6 = e\% = \frac{|N7:3 - N7:2| \times 100}{N7:2}. \quad (4)$$

The logic is enabled only when $N7:2$ is nonzero, preventing division by zero. With a 15 inHg setpoint, the nominal band widths are 0.6, 2.25, and 4.5 inHg.

Integer-resolution limitation

Because $N7:3$ and $N7:6$ are integer registers, the measured value and calculated percentage are rounded. This passes the supplied tests but makes the 4% band coarse. A real installation should calculate the scaled measurement and error in floating-point registers, then tune both the bands and the sampling period from process-response data.

3 Control Sequence



4 Complete RSLogix 500 Program

The rung order is intentional. The adjustment rungs must execute before the timer-reset rung so that T4:0/DN remains true long enough to perform exactly one correction.

4.1 Rung 000 – Scale the analog input

SCP instruction

```

Rung 000
|----[SCP Input=N7:0 InMin=0 InMax=16383
| ScaledMin=0 ScaledMax=30 Output=N7:3]----|
  
```

If SCP is unavailable, use floating-point math rather than storing the product in a 16-bit integer:

```

MUL N7:0 30.0 F8:0
DIV F8:0 16383.0 F8:0
  
```

The direct integer product $N7:0 * 30$ can exceed 32767 and should not be placed in an integer destination.

4.2 Rungs 001–004 – Calculate the error

Signed error, magnitude, and percent

```

Rung 001
|----[SUB N7:3 N7:2 N7:4]-----|

Rung 002
|----[ABS N7:4 N7:5]-----|

Rung 003
|----[NEQ N7:2 0]----[MUL N7:5 100 N7:6]-----|

Rung 004
|----[NEQ N7:2 0]----[DIV N7:6 N7:2 N7:6]-----|

```

The maximum expected engineering error is 30 inHg, so $30 * 100 = 3000$; this intermediate value fits safely in a signed 16-bit integer.

4.3 Rung 005 – Generate the ten-second interval

Periodic timer

```

Rung 005
|----[NEQ N7:2 0]-----[TON T4:0 TimeBase=1.0
| Preset=10]-----|

```

If the selected processor only offers a 0.01-second time base, use a preset of 1000.

4.4 Rungs 006–008 – Increase the VFD command

These rungs apply when $N7:4 > 0$: the measured inHg value is above the setpoint and more pump speed is required.

Positive-error correction

```

Rung 006: 4% < error <= 15%
|--[XIC T4:0/DN]--[GRT N7:4 0]--[GRT N7:6 4]
|--[LEQ N7:6 15]--[ADD N7:1 20 N7:1]-----|

Rung 007: 15% < error <= 30%
|--[XIC T4:0/DN]--[GRT N7:4 0]--[GRT N7:6 15]
|--[LEQ N7:6 30]--[ADD N7:1 60 N7:1]-----|

Rung 008: error > 30%
|--[XIC T4:0/DN]--[GRT N7:4 0]--[GRT N7:6 30]
|--[ADD N7:1 100 N7:1]-----|

```

4.5 Rungs 009–011 – Decrease the VFD command

These rungs apply when $N7:4 < 0$: the measured inHg value is below the setpoint and pump speed must be reduced.

Negative-error correction

```

Rung 009: 4% < error <= 15%
|--[XIC T4:0/DN]--[LES N7:4 0]--[GRT N7:6 4]
|--[LEQ N7:6 15]--[SUB N7:1 20 N7:1]-----|

Rung 010: 15% < error <= 30%
|--[XIC T4:0/DN]--[LES N7:4 0]--[GRT N7:6 15]
|--[LEQ N7:6 30]--[SUB N7:1 60 N7:1]-----|

Rung 011: error > 30%
|--[XIC T4:0/DN]--[LES N7:4 0]--[GRT N7:6 30]
|--[SUB N7:1 100 N7:1]-----|

```

No adjustment rung is true when $N7:6 \leq 4$; therefore, the VFD command holds its existing value inside the deadband.

4.6 Rungs 012–014 – Clamp the output and restart

Output protection and timer reset

```

Rung 012: upper clamp
|----[GRT N7:1 16383]-----[MOV 16383 N7:1]--|

Rung 013: lower clamp
|----[LES N7:1 0]-----[MOV 0 N7:1]-----|

Rung 014: restart the ten-second interval
|----[XIC T4:0/DN]-----[RES T4:0]-----|

```

Why an ONS is not required

On the scan when the timer becomes done, one correction rung executes. Rung 014 then resets the timer during that same scan, clearing T4:0/DN. The done bit therefore already behaves as a one-scan event for the correction logic.

5 Emulator Test Procedure

Set $N7:2 = 15$. Initially set $N7:0 = 8192$ and verify that $N7:3$ is approximately 15. Then set $N7:1 = 6000$; it should remain unchanged for longer than one ten-second interval.

N7:0	Approx. inHg	Expected behavior every 10 seconds
8192	15.0	Hold N7:1; error is inside the deadband
8750	16.0	Increase N7:1 by 20
7650	14.0	Decrease N7:1 by 20
10000	18.3	Increase N7:1 by 60
6200	11.4	Decrease N7:1 by 60
13000	23.8	Increase N7:1 by 100
4000	7.3	Decrease N7:1 by 100

For each test, watch T4:0/ACC, T4:0/DN, N7:3, N7:4, N7:6, and N7:1. Confirm that N7:1 changes only when the timer reaches its preset.

6 Practical Commissioning Notes

- The 10-second update interval and the specified step sizes are for emulator testing, not a recommendation for a real vacuum system.
- Begin real commissioning with conservative corrections. Trend measured vacuum and speed command, then adjust interval, bands, and step sizes based on settling time and oscillation.
- Add permissives for pump availability, VFD ready, emergency stop, maintenance mode, sensor fault, and communications health before applying the speed reference to real equipment.
- Consider separate acceleration and deceleration limits. A VFD may also provide internal ramp times that interact with the PLC's correction interval.
- In production, use actual analog output scaling appropriate to the module, such as 0–10 V or 4–20 mA, rather than assuming that 0–16383 is universally correct.

Final scan-order summary

Scale input → calculate error → run timer → execute one correction → clamp output → reset timer.
Keeping the reset rung last is essential.